c) Implementing Artificial Neural Networks (ANN)

Aim:

To build a simple neural network to classify handwritten digits (MNIST).

Procedure:

1. Load data.
2. Define a neural network.
3. Train and evaluate.

Python Code (using Keras):

```
from tensorflow.keras.datasets import mnist
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Flatten
from tensorflow.keras.utils import to_categorical

# Load data
(x_train, y_train), (x_test, y_test) = mnist.load_data()
x_train, x_test = x_train/255.0, x_test/255.0

# One-hot encode labels
y_train = to_categorical(y_train)
y_test = to_categorical(y_test)

# Define ANN
model = Sequential([
    Flatten(input_shape=(28, 28)),
    Dense(128, activation='relu'),
    Dense(10, activation='softmax')
])

# Compile & Train
model.compile(optimizer='adam', loss='categorical_crossentropy', metrics=['accuracy'])
model.fit(x_train, y_train, epochs=5, validation_split=0.1)

# Evaluate
loss, acc = model.evaluate(x_test, y_test)
print("Test Accuracy:", acc)
```

Output:

```
Downloading data from
https://storage.googleapis.com/tensorflow/tf-keras-datasets/mnist.npz
11490434/11490434 ==================== 1s 0us/step
Epoch 1/5
1688/1688 ==================== 10s 5ms/step - accuracy: 0.8642
- loss: 0.4699 - val_accuracy: 0.9648 - val_loss: 0.1215
Epoch 2/5
1688/1688 ==================== 10s 6ms/step - accuracy: 0.9611
- loss: 0.1288 - val_accuracy: 0.9690 - val_loss: 0.1027
Epoch 3/5
1688/1688 ==================== 10s 6ms/step - accuracy: 0.9755
- loss: 0.0857 - val_accuracy: 0.9747 - val_loss: 0.0867
Epoch 4/5
1688/1688 ==================== 9s 5ms/step - accuracy: 0.9822
- loss: 0.0614 - val_accuracy: 0.9755 - val_loss: 0.0840
Epoch 5/5
1688/1688 ==================== 10s 5ms/step - accuracy: 0.9871
- loss: 0.0443 - val_accuracy: 0.9765 - val_loss: 0.0797
313/313 ==================== 1s 3ms/step - accuracy: 0.9730
- loss: 0.0902
Test Accuracy: 0.9757000207901001
```

Result:

Thus the Artificial Neural Network for classifying handwritten digits (MNIST) was implemented and executed successfully.